

Problem

As a design engineer, you are asked to design a lighting system consisting of a 70-W power supply and two light bulbs as shown in Fig. 2.124. You must select the two bulbs from the following three available bulbs.

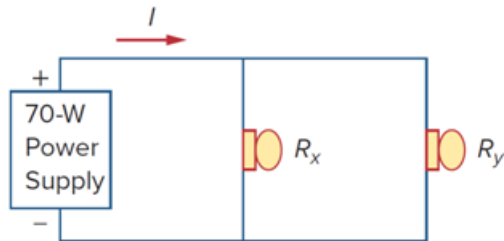
$R_1 = 80\ \Omega$, cost = \$0.60 (standard size)

$R_2 = 90\ \Omega$, cost = \$0.90 (standard size)

$R_3 = 100\ \Omega$, cost = \$0.75 (nonstandard size)

The system should be designed for minimum cost such that I lies within the range $I = 1.2\text{ A} \pm 5$ percent.

Figure 2.124



Step-by-step solution

Step 1 of 3

Consider the following design of a lighting system.

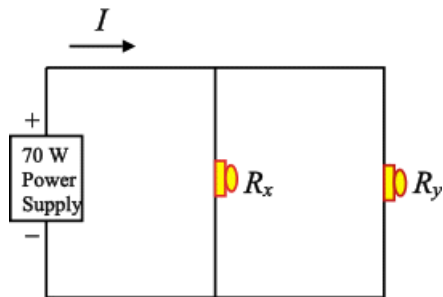


Figure 1

Step 2 of 3

Select two bulbs from the following three available bulbs:

$$R_1 = 80 \, \Omega, \text{ cost} = \$0.60 \text{ (standard size)}$$

$$R_2 = 90 \, \Omega, \text{ cost} = \$0.90 \text{ (standard size)}$$

$$R_3 = 100 \, \Omega, \text{ cost} = \$0.75 \text{ (standard size)}$$

The design should have minimum cost with the current range, $I = 1.2 \text{ A} \pm 5 \text{ percent}$.

For minimum cost, select the two bulbs, R_1 and R_3 of low cost and check whether the design constraint is met or not.

The resistances are,

$$R_x = R_1 = 80 \, \Omega$$

$$R_y = R_3 = 100 \, \Omega$$

Determine the equivalent resistance of the network.

$$\begin{aligned} R &= \frac{R_x R_y}{R_x + R_y} \\ &= \frac{(80)(100)}{80 + 100} \\ &= 44.44 \, \Omega \end{aligned}$$

Step 3 of 3

Determine the total power output of the two bulbs.

$$\begin{aligned} P &= I^2 R \\ &= 44.44 I^2 \end{aligned}$$

This power should be equal to the supply power.

$$44.44 I^2 = 70$$

$$I^2 = \frac{70}{44.44}$$

$$I = \sqrt{\frac{70}{44.44}}$$

$$I = 1.255 \text{ A}$$

As the range of current lies within the design constraint, the two bulbs selected for minimum cost specification are R_1 and R_3 .